An 18-Vertex Counterexample to the Oriented Berge–Fulkerson Conjecture

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Abstract

The oriented Berge–Fulkerson conjecture (o6c4c) asserts that every finite loopless bridgeless graph admits six oriented cycles such that every edge occurs in exactly four indexed members, twice in each direction. We construct a simple cubic 3-edge-connected counterexample on 18 vertices, obtained from the Petersen graph by replacing the four vertices of a closed neighbourhood by triangles. Any putative cover forces the six complementary perfect matchings to be the unique lifts of the six perfect matchings of the Petersen graph. A fixed integer linear combination of five edge-balance equations then yields $4z_0 = 0$ for a sign $z_0 \in \{-1, 1\}$, a contradiction. The proof is self-contained; exhaustive computation is used only as an independent verification.

1 Introduction

Ulyanov [2, Conjecture 1.1] proposed the oriented Berge–Fulkerson conjecture, abbreviated o6c4c, as an oriented strengthening of the 6-cycle 4-cover reformulation of the Berge–Fulkerson conjecture [1]. Following Ulyanov's terminology and the standard circuit-cover convention [3], a cycle is an even subgraph and need not be connected.

For the cubic graph considered below, every cycle is a disjoint union of circuits. We orient such a cycle componentwise, assigning each circuit one of its two cyclic orientations. Equivalently, the resulting directed subgraph has equal indegree and outdegree at every vertex. Thus no convention for orienting an even subgraph at a vertex of degree greater than two enters the proof.

Theorem 1.1. *There exists a finite simple cubic* 3*-edge-connected graph* H *on* 18 *vertices that admits no oriented* 6*-cycle* 4*-cover.*

This does not conflict with the computational verification for snarks on at most 30 vertices reported in [2, Section 2.3]: the graph H constructed here has girth 3 and cyclic edge-connectivity 3, and hence is outside the class considered there.

The graph is a four-triangle expansion of the Petersen graph. The argument has two rigid stages. First, local degree counting turns every putative cover into a perfect-matching double cover, and the matching structure of the Petersen graph makes that cover unique up to permuting its six indexed members. Second, direction balance on five selected edges gives an inconsistent integer system.

2 Definitions and the signed formulation

All graphs are finite and loopless. The counterexample is simple, so no convention concerning parallel edges affects the result.

Definition 2.1. A *cycle* in a graph G is an edge set $F \subseteq E(G)$ such that $d_F(v)$ is even for every $v \in V(G)$; it need not be connected. An orientation of F is *Eulerian* if

$$d_F^+(v) = d_F^-(v) \qquad (v \in V(G)).$$

An *oriented* 6*-cycle* 4*-cover*, or *o6c4c*, is an indexed six-tuple $(F_0, \dots, F_5)$ of oriented cycles such that every edge occurs in exactly four members and, among those four occurrences, is directed twice in each direction. Repetitions among the underlying edge sets are permitted.

When G is cubic, every cycle has degree 0 or 2 at each vertex. Hence its nontrivial connected components are circuits, and an Eulerian orientation is precisely a choice of one of the two cyclic orientations on each circuit component.

Fix a reference direction for every edge of G. An oriented cycle F_i is encoded by a function

$$s_i \colon E(G) \longrightarrow \{-1, 0, 1\},$$

where $s_i(e) = 0$ if $e \notin F_i$, $s_i(e) = 1$ if the orientation of e in F_i agrees with the reference direction, and $s_i(e) = -1$ otherwise. Let $\varepsilon(v, e) = 1$ if the reference direction of e points away from v, let $\varepsilon(v, e) = -1$ if it points towards v, and let $\varepsilon(v, e) = 0$ if e is not incident with v.

Lemma 2.2 (Signed formulation)**.** *A graph G has an oriented* 6*-cycle* 4*-cover if and only if there are functions*

$$s_0, \dots, s_5 \colon E(G) \longrightarrow \{-1, 0, 1\}$$

satisfying

$$\sum_{e \in E(G)} \varepsilon(v, e) s_i(e) = 0 \qquad (v \in V(G),\ 0 \le i \le 5), \tag{1}$$

$$\sum_{i=0}^{5} |s_i(e)| = 4 \qquad (e \in E(G)), \tag{2}$$

$$\sum_{i=0}^{5} s_i(e) = 0 \qquad (e \in E(G)). \tag{3}$$

Proof. Equation (1) is precisely vertex balance in each member, while (2) says that four indexed members contain the edge. Four nonzero signs in $\{-1, 1\}$ have sum zero exactly when two are positive and two are negative, giving (3). The converse reconstruction is immediate.

Reversing the reference direction of an edge negates both $\varepsilon(\,\cdot\,, e)$ and all six values $s_i(e)$. Thus (1) is unchanged, while (2) and (3) remain equivalent to their original forms. □

3 The 18-vertex graph

Let P be the Petersen graph on vertices $0, \dots, 9$ with edge set

$$E(P) = \{01, 04, 05, 12, 16, 23, 27, 34, 38, 49, 57, 58, 68, 69, 79\}. \tag{4}$$

Here and below, vw denotes the edge $\{v, w\}$. Set $X = N_P[0] = \{0, 1, 4, 5\}$. For each $v \in X$, replace v by a triangle whose three vertices are ports p_{vw} indexed by the neighbours w of v. Join the three ports belonging to the same expanded vertex in a triangle, and denote an unexpanded vertex v by u_v. For an old edge $vw \in E(P)$, its image in H is

$$\begin{cases} p_{vw}p_{wv}, & v, w \in X, \\ p_{vw}u_w, & v \in X,\ w \notin X, \\ u_v p_{wv}, & v \notin X,\ w \in X, \\ u_v u_w, & v, w \notin X. \end{cases}$$

Denote the resulting graph by H.

Lemma 3.1. *Replacing a vertex of a cubic* 3*-edge-connected graph by a triangle, with its three old incident edges attached to distinct triangle vertices, preserves* 3*-edge-connectivity.*

Proof. Suppose the expanded graph has an edge cut of size at most two. If the cut does not split the new triangle, contracting that triangle gives a cut of the same size in the original graph. If the cut splits the triangle, two internal triangle edges already cross it. No other edge can cross. Each external neighbour is therefore on the same side as its port. Contract the triangle and place the old vertex on the side of the lone port. The two external edges belonging to the other ports form a cut of size two in the original graph, a contradiction. □

Proposition 3.2. *The graph H is simple, cubic,* 3*-edge-connected, and has* 18 *vertices and* 27 *edges. It has girth* 3 *and cyclic edge-connectivity* 3*.*

Proof. The construction is simple and cubic. Replacing four vertices adds eight vertices and twelve edges, so $|V(H)| = 10 + 8 = 18$ and $|E(H)| = 15 + 12 = 27$.

For completeness, P is 3-edge-connected. It is cubic and triangle-free. For a nontrivial $S \subset V(P)$, pass to the smaller shore so that $1 \le |S| \le 5$. Since

$$|\delta(S)| = 3|S| - 2|E(P[S])|,$$

the triangle-free bounds

$$|E(P[S])| \le 0, 1, 2, 4, 6 \quad \text{for} \quad |S| = 1, 2, 3, 4, 5$$

give $|\delta(S)| \ge 3$ in every case. Apply Lemma 3.1 four times.

Each replacement triangle shows that the girth is 3. Its three external edges form a cyclic edge cut: one shore contains the triangle, and the other shore contains a cycle. Thus the cyclic edge-connectivity is at most 3, and 3-edge-connectivity gives the reverse inequality. □

An exact numbered edge list for H appears in Appendix A.

4 Rigidity of an unoriented cover

Suppose that $F_0, \ldots, F_5$ are cycles of H that cover every edge exactly four times; no orientations are assumed in this section. Set

$$N_i = E(H) \setminus F_i \qquad (0 \le i \le 5).$$

Lemma 4.1 (Support saturation). *Every F_i is a spanning 2-factor. Consequently every N_i is a perfect matching, and $(N_0, \dots, N_5)$ covers every edge exactly twice.*

Proof. Since F_i is a cycle, $d_{F_i}(v)$ is even for every vertex v. Since H is cubic,

$$d_{F_i}(v) \in \{0, 2\}.$$

On the other hand, each of the three edges incident with v belongs to exactly four of the six supports. Therefore

$$\sum_{i=0}^{5} d_{F_i}(v) = \sum_{e \ni v} \sum_{i=0}^{5} \mathbf{1}_{e \in F_i} = 3 \cdot 4 = 12.$$

The six summands are each at most 2, so all of them are equal to 2. Thus every F_i is a spanning 2-factor.

Consequently, $N_i = E(H) \setminus F_i$ meets every vertex exactly once and is therefore a perfect matching. Since each edge lies in four of the supports, it lies in exactly two of the complements. $\square$

The Petersen graph has exactly the following six perfect matchings:

$$\begin{aligned} Q_0 &= \{01, 23, 49, 57, 68\}, & Q_1 &= \{01, 27, 34, 58, 69\}, \\ Q_2 &= \{04, 12, 38, 57, 69\}, & Q_3 &= \{04, 16, 23, 58, 79\}, \\ Q_4 &= \{05, 12, 34, 68, 79\}, & Q_5 &= \{05, 16, 27, 38, 49\}. \end{aligned} \tag{5}$$

Indeed, branch on the matching edge incident with vertex 0. Each of $01, 04, 05$ leaves precisely the two completions displayed above. Direct comparison gives

$$|Q_j \cap Q_k| = 1 \quad (j \neq k), \qquad |Q_j| = 5. \tag{6}$$

The table also shows that every edge of P belongs to exactly two of the six matchings. For $j = 0, \dots, 5$, define the lift $\widetilde{Q}_j$ by using the image in H of every edge of Q_j and, at each replacement triangle, the internal edge joining the two ports not incident with the selected external edge. Since each Petersen edge occurs in two of the Q_j, and each internal triangle edge is selected by the two matchings using the opposite external port, the six perfect matchings

$$\widetilde{Q}_0, \dots, \widetilde{Q}_5$$

double-cover every edge of H.

Lemma 4.2 (Unique matching cover). *After permuting the six indices,*

$$(N_0, \dots, N_5) = (\widetilde{Q}_0, \dots, \widetilde{Q}_5).$$

Proof. Fix one replacement triangle. A perfect matching uses at most one internal edge there. The three internal edges have total multiplicity six among the six matchings N_i, because every edge belongs to exactly two of them. Hence each N_i uses exactly one internal edge and therefore exactly one external edge at this triangle. This holds at all four triangles. Contracting the triangles maps every N_i to a perfect matching of P, and the construction above shows that its lift is unique.

Let y_j be the multiplicity of Q_j among the six contracted matchings. Then $\sum_{j=0}^{5} y_j = 6$. For fixed k, sum the double-cover equations over the five edges of Q_k. By (6),

$$\begin{aligned} 10 &= \sum_{j=0}^{5} y_j |Q_k \cap Q_j| \\ &= 5y_k + \sum_{j \neq k} y_j \\ &= 5y_k + (6 - y_k) \\ &= 6 + 4y_k. \end{aligned}$$

Thus $y_k = 1$ for every $k = 0, \ldots, 5$. □

Corollary 4.3. *Up to a permutation of its six members, H has a unique unoriented 6-cycle 4-cover, namely*

$$H - \widetilde{Q}_0, \ldots, H - \widetilde{Q}_5.$$

Proof. Existence follows because the six lifted matchings double-cover every edge, so their complements cover every edge four times. Uniqueness follows from Lemmas 4.1 and 4.2. □

5 The directional obstruction

Use the vertex order

$$\begin{aligned} &p_{01}, p_{04}, p_{05}, p_{10}, p_{12}, p_{16}, u_2, u_3, p_{40}, p_{43}, p_{49}, \\ &\qquad p_{50}, p_{57}, p_{58}, u_6, u_7, u_8, u_9 \end{aligned} \tag{7}$$

and direct every edge from the earlier endpoint to the later endpoint as the reference direction.

For each j, the complement $H - \widetilde{Q}_j$ is the disjoint union of an 11-circuit D_j and a 7-circuit D'_j. The component D_j containing p_{01} has the following cyclic order:

$$\begin{aligned} D_0 &= (p_{01}, p_{04}, p_{40}, p_{49}, p_{43}, u_3, u_8, p_{58}, p_{57}, p_{50}, p_{05}), \\ D_1 &= (p_{01}, p_{04}, p_{40}, p_{43}, p_{49}, u_9, u_7, p_{57}, p_{58}, p_{50}, p_{05}), \\ D_2 &= (p_{01}, p_{04}, p_{05}, p_{50}, p_{57}, p_{58}, u_8, u_6, p_{16}, p_{12}, p_{10}), \\ D_3 &= (p_{01}, p_{04}, p_{05}, p_{50}, p_{58}, p_{57}, u_7, u_2, p_{12}, p_{16}, p_{10}), \\ D_4 &= (p_{01}, p_{05}, p_{04}, p_{40}, p_{43}, p_{49}, u_9, u_6, p_{16}, p_{12}, p_{10}), \\ D_5 &= (p_{01}, p_{05}, p_{04}, p_{40}, p_{49}, p_{43}, u_3, u_2, p_{12}, p_{16}, p_{10}). \end{aligned}$$

The six circuits D'_j are listed in Appendix B.

Consider the five reference-directed edges

$$e_1 = p_{01}p_{04}, \quad e_2 = p_{05}p_{50}, \quad e_3 = p_{12}p_{16}, \quad e_4 = p_{43}p_{49}, \quad e_5 = p_{57}p_{58}. \tag{8}$$

A direct check shows that, for every $r \in \{1, \ldots, 5\}$ and $j \in \{0, \ldots, 5\}$, either $e_r \in \widetilde{Q}_j$ or $e_r \in E(D_j)$. In particular, none of the five selected edges belongs to D'_j. Consequently, the orientation variables of the six 7-circuits do not occur in the five balance equations below.

Let $z_j \in \{-1,1\}$ record whether the orientation on D_j agrees with the displayed cyclic order. Reading the directions of the five selected edges in the six displayed circuits, equation (3) gives

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 & 1 & -1 \\ -1 & 1 & 1 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} z_0 \\ z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \end{pmatrix} = 0. \tag{9}$$

Lemma 5.1 (Five-row obstruction)**.** *The system* (9) *has no solution in* $\{-1,1\}^6$.

Proof. Let $r_1, \dots, r_5$ be the rows of the matrix in (9). Then

$$r_1 - r_2 - r_3 - r_4 - r_5 = (4,0,0,0,0,0).$$

Left-multiplying (9) by $(1,-1,-1,-1,-1)$ therefore gives $4z_0 = 0$, contradicting $z_0 \in \{-1,1\}$. □

Proof of Theorem 1.1. By Proposition 3.2, H is a finite simple cubic 3-edge-connected graph. If it admitted an oriented 6-cycle 4-cover, Lemma 4.1 would turn the six complementary edge sets into a perfect-matching double cover. By Lemma 4.2, after permuting the indices these matchings would be $\widetilde{Q}_0, \dots, \widetilde{Q}_5$. The orientations of the corresponding 2-factors would then produce signs satisfying (9), contrary to Lemma 5.1. □

6 Independent computational verification

The proof above is self-contained and does not use computation as a premise. For reproducibility, two separately written standard-library Python implementations verify the finite data. The first constructs H from the Petersen triangle expansion and derives the orientation equations from its circuit decomposition. The second starts from a separately numbered edge list and independently enumerates the same structures. The source code, exact input files, expected output, and reproduction instructions are packaged in the fixed release `o6c4c-counterexample-v1.0.0.zip`, whose SHA-256 digest is

`2243f4e96bc58f28c386a5b57f8af5a63ad43ebbf9efa323627f15be757b9d71`.

The fixed release is publicly available at `https://github.com/Miracleofwinds/o6c4c-counterexample/releases/tag/v1.0.0`.

The verification includes:

- connectivity after all $1 + 27 + \binom{27}{2} = 379$ deletions of at most two edges;
- the six Petersen perfect matchings and all 18 perfect matchings of H;
- the unique double cover among 100,947 multisets of six perfect matchings;
- all twelve circuit components and all signs in (9); and
- infeasibility of all $2^{12} = 4{,}096$ component-orientation assignments.

7 Scope of the counterexample

Because H is cubic, every cycle in H is a disjoint union of circuits. Thus the proof uses only the componentwise cyclic orientation of ordinary circuits; no convention concerning orientations at vertices of degree greater than two is involved.

Disconnected cycles and repetitions among the six indexed members are permitted throughout. Consequently, the nonexistence result also applies to any stronger formulation requiring the six underlying edge sets to be distinct.

The graph H does admit an unoriented 6-cycle 4-cover, uniquely up to permutation, by Corollary 4.3. The obstruction is therefore genuinely orientational. Finally, H contains triangles and has cyclic edge-connectivity 3, so it lies outside the class of snarks of girth at least 5 and cyclic edge-connectivity at least 4 tested computationally in [2, Section 2.3].

A Exact numbered graph data

For direct reproduction, number the vertices in the order (7):

$$\begin{array}{rcl rcl rcl}
0 & : & p_{01} & 1 & : & p_{04} & 2 & : & p_{05} \\
3 & : & p_{10} & 4 & : & p_{12} & 5 & : & p_{16} \\
6 & : & u_2 & 7 & : & u_3 & 8 & : & p_{40} \\
9 & : & p_{43} & 10 & : & p_{49} & 11 & : & p_{50} \\
12 & : & p_{57} & 13 & : & p_{58} & 14 & : & u_6 \\
15 & : & u_7 & 16 & : & u_8 & 17 & : & u_9
\end{array}$$

The 27 edges, each listed in its reference direction, are

$$\begin{aligned}
&(0,3),(1,8),(2,11),(4,6),(5,14),(6,7),(6,15),(7,9),(7,16),\\
&(10,17),(12,15),(13,16),(14,16),(14,17),(15,17),\\
&(0,1),(0,2),(1,2),(3,4),(3,5),(4,5),\\
&(8,9),(8,10),(9,10),(11,12),(11,13),(12,13).
\end{aligned}$$

The first fifteen edges correspond to the Petersen edges in (4); the remaining twelve are the edges of the four replacement triangles.

B Complete circuit decompositions

For completeness, the second component of $H - \widetilde{Q}_j$ is the following 7-circuit:

$$\begin{aligned}
D'_0 &= (p_{10}, p_{12}, u_2, u_7, u_9, u_6, p_{16}),\\
D'_1 &= (p_{10}, p_{12}, u_2, u_3, u_8, u_6, p_{16}),\\
D'_2 &= (u_2, u_7, u_9, p_{49}, p_{40}, p_{43}, u_3),\\
D'_3 &= (u_3, p_{43}, p_{40}, p_{49}, u_9, u_6, u_8),\\
D'_4 &= (u_2, u_3, u_8, p_{58}, p_{50}, p_{57}, u_7),\\
D'_5 &= (p_{50}, p_{58}, u_8, u_6, u_9, u_7, p_{57}).
\end{aligned}$$

Together with the six D_j in Section 5, these are all twelve components of the six complementary 2-factors.

C A minimal manual audit of the matrix

To check (9) without a program, use the five edges in (8). For a fixed row and column, enter 0 if the edge belongs to $\widetilde{Q}_j$, $+1$ if the cyclic sequence for D_j traverses it in the reference direction, and -1 otherwise. This gives the displayed matrix entry by entry. Its row combination is

$$(1,-1,-1,-1,-1)\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 & 1 & -1 \\ -1 & 1 & 1 & -1 & 0 & 0 \end{pmatrix} = (4,0,0,0,0,0),$$

which is the complete final obstruction.

Acknowledgements

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References

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